

Reading the Atmosphere Profile

The Live Map Skew-T diagram of MODE-S Wind

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In short. The Atmosphere Profile panel on the right of the Live Map is a small *Skew-T log-p* diagram: a vertical profile of temperature and wind, like a weather balloon sounding. Its data do not come from a balloon but from the aircraft themselves. Their transponders report airspeed, Mach number, heading and track, and from these MODE-S Wind computes the air temperature and the wind around each aircraft. The grey curve is the average of all aircraft within 150 km over the last 60 minutes; clicking an aircraft adds its own profile in colour. An Analysis button marks the tropopause, the freezing level, jet streams, inversions, stability, turbulence risk, warm and cold advection, contrail levels and the daytime thermal top. There is no humidity in the data, so cloud and moisture cannot be shown.

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1 What the panel shows and where

The panel fills the right-hand side of the Live Map page (Fig. 1). It always shows the **area profile**: all aircraft within 150 km of the receiver over the last 60 minutes, refreshed every 5 minutes. When you click an aircraft on the map or in the list, its own profile is drawn on top in colour; click an empty spot on the map, press Esc or close the aircraft's popup to return to the area profile.

Under the diagram are two controls, Density (how many points of a selected aircraft are drawn) and Smooth (vertical smoothing of the area temperature profile). In the panel title is the Analysis button, which steps through six levels of automatic analysis (Section 8). Hovering over the diagram shows the values of the nearest level.

A Skew-T is read from bottom to top. Height is not drawn in metres but as *air pressure*, which falls with height; temperature lines are tilted (“skewed”) to the right. This looks unusual at first, but it makes the important features of the atmosphere stand out: how fast temperature falls with height, where it suddenly stops falling, and where the air is stable or unstable.

2 Upper-air parameters in brief

Pressure (hPa). The weight of the air above. About 1013 hPa at sea level, 850 hPa at about 1.5 km, 700 hPa at 3 km, 500 hPa at 5.5 km, 300 hPa at 9 km and 250 hPa at 10.4 km. Meteorologists use pressure as the vertical coordinate because weather maps and soundings are made at fixed pressure levels.

Pressure altitude and flight level (FL). The altitude an aircraft reports is its *pressure altitude*: the height in the International Standard Atmosphere (ISA) at which the measured pressure would occur, always referenced to 1013.25 hPa. FL350 means 35 000 ft pressure altitude. Pressure altitude is therefore a direct measure of pressure, not of true height above the sea.

True altitude. The real height. It differs from pressure altitude by about 27 ft for each hPa that the sea-level pressure (QNH) differs from 1013, and further in air warmer or colder than ISA (colder air is denser, so the same pressure lies lower).

Temperature and ISA. The ISA reference falls by 6.5 °C per km (about 2 °C per 1 000 ft) from +15 °C at sea level to −56.5 °C at 11 km and stays constant above. Real air is warmer or colder; the difference is the *ISA deviation*.

Lapse rate. How fast temperature falls with height, in K/km. 6.5 K/km is the ISA average; 9.8 K/km is the *dry adiabatic* rate at which unsaturated rising air cools; 4–7 K/km is the *moist adiabatic* rate of saturated air (slower, because condensing water releases heat).

Stability. If the air’s lapse rate is smaller than that of a rising air parcel, the parcel becomes colder than its surroundings and sinks back: the layer is *stable*. If larger, the parcel keeps rising: *unstable*. Between the dry and moist rates the layer is *conditionally unstable*: stable for clear air, unstable once the air is saturated (showers, cumulus).

Inversion. A layer where temperature *rises* with height. Very stable; it traps haze, fog and low cloud below it, and often forms near the ground on clear nights.

Freezing level. The height of the 0 °C isotherm. With a warm layer aloft there can be more than one.

Tropopause. The top of the weather layer (troposphere), where temperature stops falling with height; typically 8–12 km over Finland, lower in cold air and lows, higher in warm air.

Wind. Direction the wind blows *from* (degrees true) and speed in knots. The strongest winds, the *jet stream*, are usually just below the tropopause.

Wind shear. Change of wind with height. Strong shear in stable air can still overturn and cause clear-air turbulence.

Advection. Transport of warmer or colder air by the wind. It changes the profile over the following hours.

Humidity, dewpoint. The amount of water vapour. A radiosonde measures it; aircraft transponders do not report it, so it is **not available** in MODE-S Wind (Section 9).

3 Where the data come from

MODE-S Wind listens to the 1090 MHz replies of aircraft transponders. Many airliners answer ground radar interrogations with *Enhanced Surveillance* (EHS) reports:

- **BDS 5,0** (track and turn): true track, groundspeed, true airspeed (TAS), roll angle;
- **BDS 6,0** (heading and speed): magnetic heading, Mach number, indicated airspeed, vertical rate;
- **BDS 4,4** (meteorological routine air report, MRAR): wind and temperature measured on board — reported by only a few aircraft;
- and the aircraft’s **ADS-B** broadcasts with position, altitude and velocity.

Temperature from Mach and TAS. The air-data computer measures both the true airspeed and the Mach number. Their ratio is the speed of sound, which depends only on the air temperature:

$$T \text{ [K]} = 288.15 \left(\frac{\text{TAS}}{661.47 \text{ kt} \times M} \right)^2.$$

A single reply pair scatters by about ±1.3 K at cruise and ±3–4 K on approach, because Mach is transmitted in steps of 0.004. Layer averages of many replies are much better: compared with the Jokioinen radiosonde (8 October 2026, 12 UTC) the mean difference was +0.7 K with a standard deviation of 1.6 K. The rare MRAR temperature is used first when available.

Wind from the wind triangle. The aircraft moves over the ground with its *ground vector* (track and groundspeed) and

through the air with its *air vector* (true heading and TAS). The difference is the wind:

$$\vec{V}_{\text{wind}} = \vec{V}_{\text{ground}} - \vec{V}_{\text{air}}.$$

The heading is corrected for magnetic declination, and replies taken during turns are rejected. MRAR winds are preferred when reported.

Pressure. Each observation's pressure is the ISA pressure of its pressure altitude — which, by the definition of pressure altitude, is the actually measured static pressure.

The area profile. Observations from the last 60 minutes within 150 km of the receiver are grouped into layers 1 000 ft thick up to 10 000 ft and 2 000 ft thick above. For each layer:

- **temperature** = 10 % trimmed mean (the warmest and coldest 10 % removed), with the 10–90 % range and the number of replies;
- **wind** = vector mean of all winds in the layer, with its scatter (SD) and number.

A typical hour gives 15–25 layers with several hundred observations at cruise levels, fewer low down and very few at night.

4 The diagram, element by element

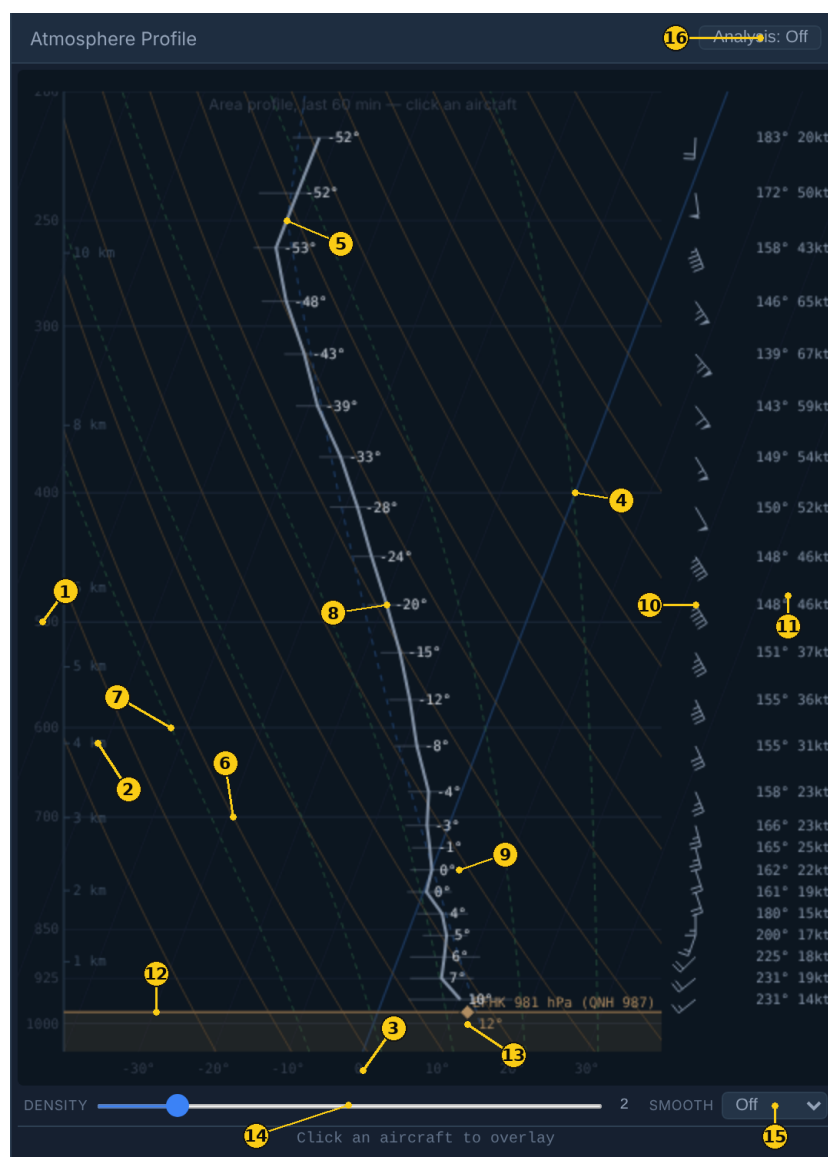
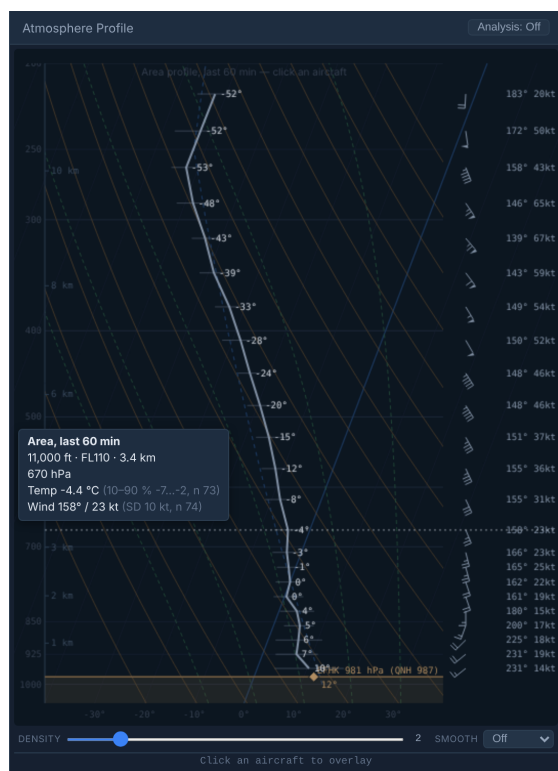


Figure 1. The Atmosphere Profile panel with the area profile of 9 October 2026, 11–12 UTC (Analysis off). The numbers refer to the list in Section 4. The surface point uses an illustrative METAR (QNH 987 hPa, 12 °C).

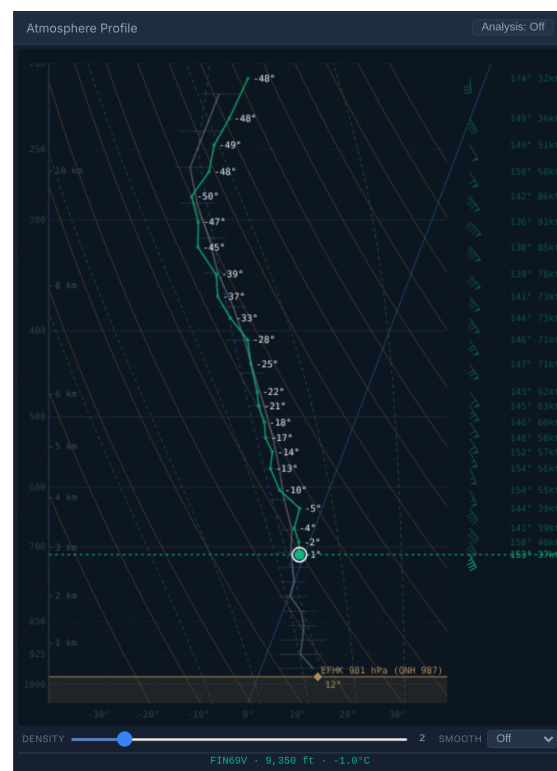
(1) **Pressure axis** (hPa), logarithmic: equal distances mean equal pressure *ratios*, which is close to equal height steps.

- (2) **Height markers** (1–12 km), pressure altitude. Use them for orientation; the true height differs a little (Section 2).
- (3) **Temperature scale** (−30 ... +30 °C), valid along the bottom of the plot. Higher up you must follow the tilted isotherms.
- (4) **Isotherms**: lines of equal temperature, tilted 45°-like to the right; the 0 °C isotherm is drawn stronger (blue). A temperature is read by following its isotherm, not by looking straight down.
- (5) **ISA reference** (dashed blue): the standard-atmosphere temperature. Curve left of it = colder than ISA, right = warmer. On this day the measured profile is almost on top of it.
- (6) **Dry adiabats** (faint solid brown, every 10 K): the path of rising or sinking unsaturated air.
- (7) **Moist adiabats** (faint dashed green): the path of rising saturated air. They are steeper than the dry adiabats near the ground and become parallel to them high up, where the air holds almost no water.
- (8) **Area temperature profile** (grey line) with the layer values in °C.
- (9) **10–90 % range** of each layer (thin horizontal bar): a wide bar means the aircraft disagreed, because of few replies, low speed (approach) or real differences across the 150 km area.
- (10) **Wind barbs**: the staff points into the wind direction; a half barb is 5 kt, a full barb 10 kt, a pennant 50 kt.
- (11) **Wind values**: direction (° from) and speed (kt) of each layer.
- (12) **Ground line** (brown, with shading below): the station pressure of EFHK from the current METAR QNH and the airport elevation (about QNH − 6.5 hPa). The label reads e.g. EFHK 981 hPa (QNH 987). Everything below the line is underground: with a low QNH the lowest pressure-altitude layer may lie below the surface and is then empty.
- (13) **Surface temperature** (diamond) from the METAR.
- (14) **Density slider**: thins the points drawn for a selected aircraft (Section 6).
- (15) **Smooth selector**: vertical smoothing of the area profile (Section 7).
- (16) **Analysis button**: automatic analysis, six levels (Section 8).

Hover read-out. Moving the mouse over the diagram shows the nearest level (Fig. 2a): its pressure altitude in ft, as a flight level and in km, the pressure, the temperature with its 10–90 % range and number of replies, and the wind with its scatter (SD) and number. A wide range or a small number tells you how far to trust the value. Hovering the ground line shows the METAR time, QNH, station pressure and temperature.



(a) Hover read-out of the 11 000 ft layer: 670 hPa, −4.4 °C (10–90 % −7...−2 °C, 73 replies), wind 158°/23 kt (SD 10 kt, 74 winds).



(b) FIN69V descending from FL390 towards EFHK (green) over the grey area profile; the circle is its present position (9 350 ft, −1 °C) with a dotted line to its current wind.

Figure 2. Hover read-out and a selected aircraft (9 October 2026, 11–12 UTC).

5 Reading the profile

1. **Start at the ground line** and the surface temperature. Compare the lowest layers with it: if the air just above is *warmer* than the surface, there is a surface inversion (typical on clear nights and mornings).
2. **Follow the curve upwards** and compare its slope with the adiabats:
 - steeper (leaning more to the left) than the dry adiabats: *unstable* — rare except right above sun-heated ground;
 - between the dry and the moist adiabats: *conditionally unstable* — showers and cumulus are possible if the air is moist enough;
 - less steep than the moist adiabats: *stable*;
 - vertical along an isotherm or leaning right: *isothermal layer or inversion* — very stable, a lid for haze, fog and low cloud.
3. **Find the freezing level** where the curve crosses the bold 0 °C isotherm, and the −10 and −20 °C isotherms. Between 0 and −20 °C icing is possible *if* the aircraft is in cloud.
4. **Find the tropopause**: the bend where the curve turns from sloping to vertical or even to the right. Above it, in the stratosphere, temperature stays nearly constant.
5. **Read the winds**: speed usually increases towards the tropopause (jet stream) and decreases above it. Wind *turning* with height tells about advection: clockwise (veering) means warm air arriving, anticlockwise (backing) cold air.
6. **Compare with ISA**: a profile well left of the dashed blue line is a cold air mass, right of it a warm one. Aircraft performance (TAS, climb rate) depends on this.

Example (Fig. 1). On 9 October 2026 at 11–12 UTC a deep low (QNH 987 hPa) lay over southern Finland. The profile stays close to ISA all the way up. Just above the ground the temperature falls quickly (10 °C at 1 500 ft, 7 °C at 2 500 ft); between 2 500 and 4 500 ft it falls only slowly, a stable layer; at 6 500–7 500 ft it stays near 0 °C, so the freezing level (FL073) lies in an almost isothermal layer. Above, the curve follows the moist adiabats fairly closely, as is typical of a saturated low. It bends at about 260 hPa: the tropopause is at FL330 and −52 °C, low because of the low. A south-easterly jet of 67 kt blows at FL290. The winds *back* from 230° near the ground to 160° at 6 500 ft — cold air arriving at low levels (Fig. 5b).

6 A selected aircraft

Clicking an aircraft draws its own profile in its map colour (Fig. 2b). It combines the observations stored for the current flight (loaded when you select it) with new live data, so a descending arrival builds a complete profile from cruise to the runway. The circle marks the latest point; its values are shown in the line under the diagram. Wind barbs and values on the right then belong to the aircraft, and the area profile stays in the background in grey.

A single aircraft is noisier than the area average, especially at low speed: compare it with the grey curve rather than reading single points. The Density slider (1–8) draws a point only every 400× density ft of altitude change, to keep the labels readable. The aircraft's track on the map can be shown with the Track checkbox.

7 Smoothing

The Smooth selector applies a vertical running mean to the temperatures of the area profile: 3 layers (weights 1-2-1) or 5 layers (1-2-3-2-1), each layer also weighted by the square root of its number of replies. Smoothing removes layer-to-layer jitter and makes it easier to compare the slope of the curve with the adiabats (Fig. 3). It is for display only: the hover read-out then shows both the smoothed value and the layer's own value, and the 10–90 % bars stay unsmoothed. Strong smoothing also softens real features — inversions and the tropopause — so switch it off when you look at those. The automatic analysis always uses its own light smoothing, whatever this selector shows.

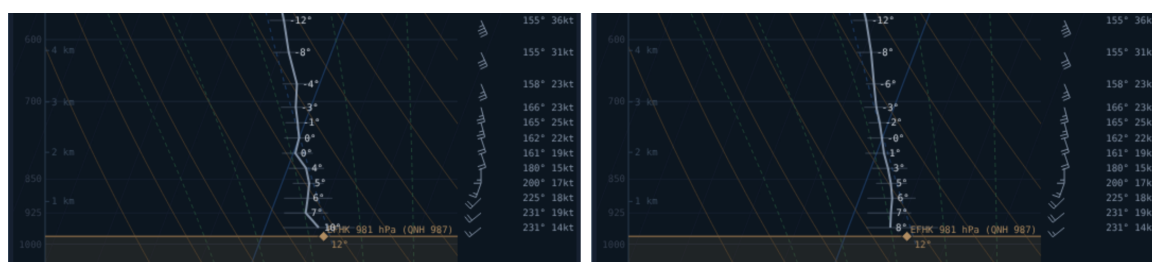


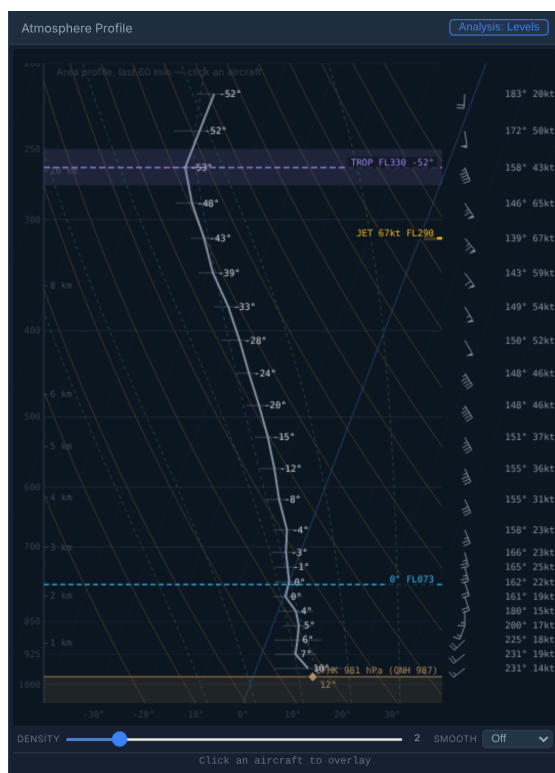
Figure 3. The lowest 4 km of the profile of Fig. 1 without smoothing (left) and with 5 layers (right).

8 The Analysis button

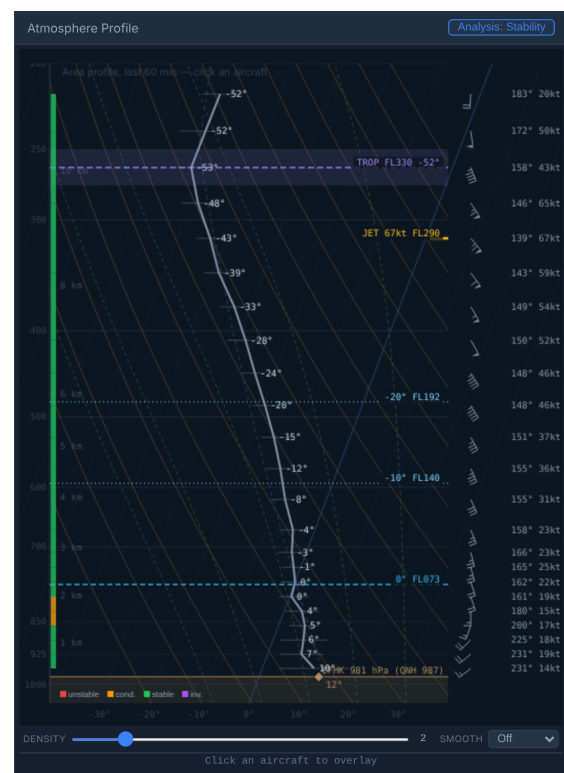
The Analysis button at the right of the panel title runs an automatic analysis of the 60-minute area profile. Each click (or a turn of the mouse wheel over the button) steps to the next level; each level adds to the previous ones. Lines and bands are drawn across the plot, labels at its right edge (spread so that they do not overlap), and marker bars along the right edge and next to the pressure axis.

Table 1. Analysis levels.

Level	Adds
Off	nothing
Levels	tropopause, 0 °C level(s), maximum wind / jet stream, low-level jet
Stability	−10 and −20 °C levels, inversions, stability strip
Turbulence	wind shear and turbulence risk (Richardson number)
Advection	warm and cold air advection, contrail levels
Full	daytime thermal top



(a) Levels: tropopause band (violet) TROP FL330 −52°, jet JET 67kt FL290 (yellow tick at the right edge), freezing level 0° FL073 (dashed blue).



(b) Stability: adds the −10 / −20 °C levels (dotted) and the stability strip next to the pressure axis (key at the bottom left).

Figure 4. Analysis levels 1 and 2, 9 October 2026, 11–12 UTC.

8.1 Levels

Tropopause (violet band $\pm 1\,000$ ft and dashed line, e.g. TROP FL330 −52°). Found with the WMO definition: the lowest level above 5 km where the lapse rate falls to 2 K/km or less and stays so, on average, over the next 2 km. A ? after the label means there were few data above it, so the level is uncertain. At night, with little high-level traffic, it may not be found at all.

0 °C level (dashed blue, e.g. 0° FL073). Every crossing of 0 °C is marked; a second or third crossing appears only when a warm or cold layer really differs by at least 1 K from 0 °C — small wiggles around zero are ignored.

Max wind / jet (yellow tick at the right edge): the strongest layer wind, labelled JET when it is 60 kt or more (the WMO jet-stream criterion), otherwise MAX.

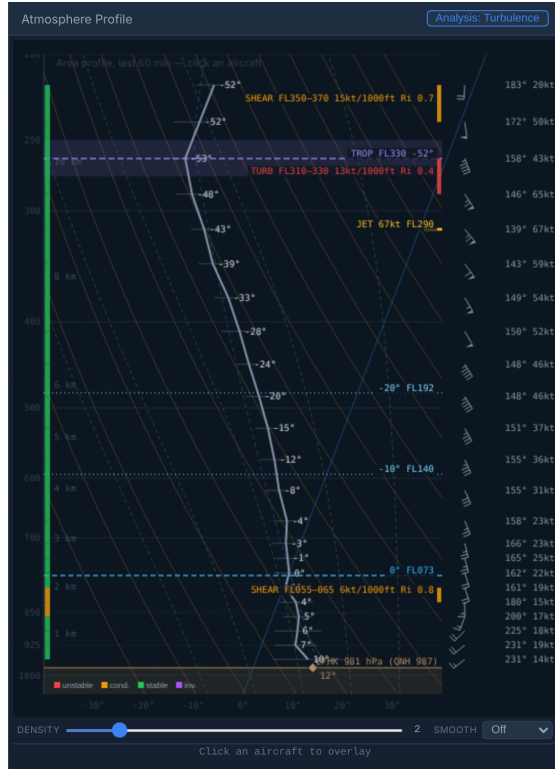
Low-level jet (LLJ): a wind maximum below 5 000 ft of at least 25 kt that is at least 10 kt stronger than the wind within 4 000 ft above it. Typical at night over a cooled surface; gives low-level wind shear on approach.

8.2 Stability

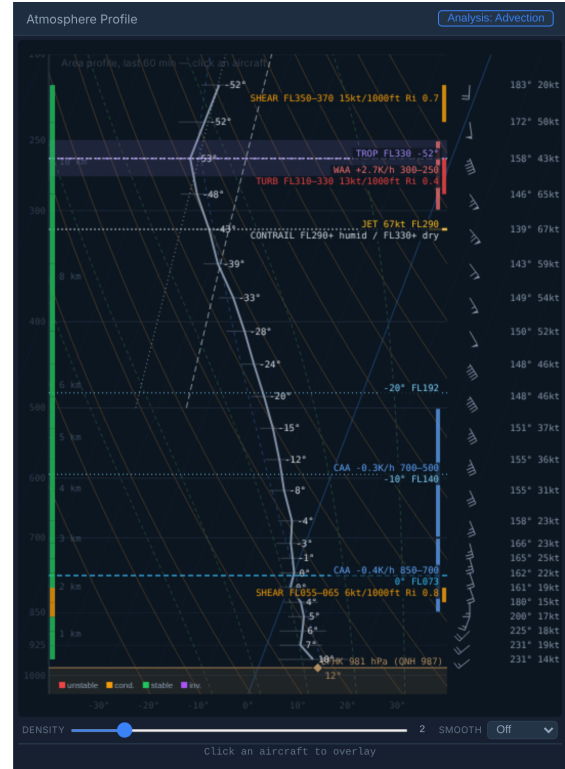
−10 and −20 °C levels (dotted). Together with 0 °C they bound the temperature range where airframe icing occurs *in cloud*. Whether there is cloud cannot be told from these data.

Inversions (orange shading, e.g. INV +2.5° FL035–045): layers where the temperature rises with height by more than twice the uncertainty of the two layer means (at least 1 K). Only below the tropopause — the stratosphere above is an inversion anyway (Fig. 6b).

Stability strip (narrow coloured bar right of the pressure axis): each layer's lapse rate compared with the adiabats — **red** absolutely unstable (> 9.8 K/km), **amber** conditionally unstable (between the moist and the dry rate), **green** stable, **violet** inversion. Above the tropopause it is usually violet.



(a) Turbulence: TURB FL310–330 (red bar) at the tropopause below the jet, SHEAR (amber) at FL350–370 and FL055–065.



(b) Advection: cold advection CAA (blue bar) at 850–500 hPa, warm WAA at 300–250 hPa; contrail threshold curves (dashed / dotted) and CONTRAIL FL290+ humid / FL330+ dry.

Figure 5. Analysis levels 3 and 4, 9 October 2026, 11–12 UTC.

8.3 Turbulence

For each pair of adjacent wind layers the **vertical wind shear** (vector difference, in kt per 1 000 ft) and the bulk **Richardson number**

$$Ri = \frac{N^2}{(\Delta V / \Delta z)^2}, \quad N^2 = \frac{g}{\theta} \frac{\Delta \theta}{\Delta z}$$

are computed, where θ is the potential temperature. Ri compares the stability, which damps vertical motion, with the shear, which drives it: below about 0.25 the flow tends to overturn into turbulence; up to 1 it is marginal.

- TURB (red bar and label): $Ri < 0.5$ and at least 4 kt per 1 000 ft;
- SHEAR (amber): $Ri < 1$ with at least 4 kt per 1 000 ft, or shear of at least 8 kt per 1 000 ft.

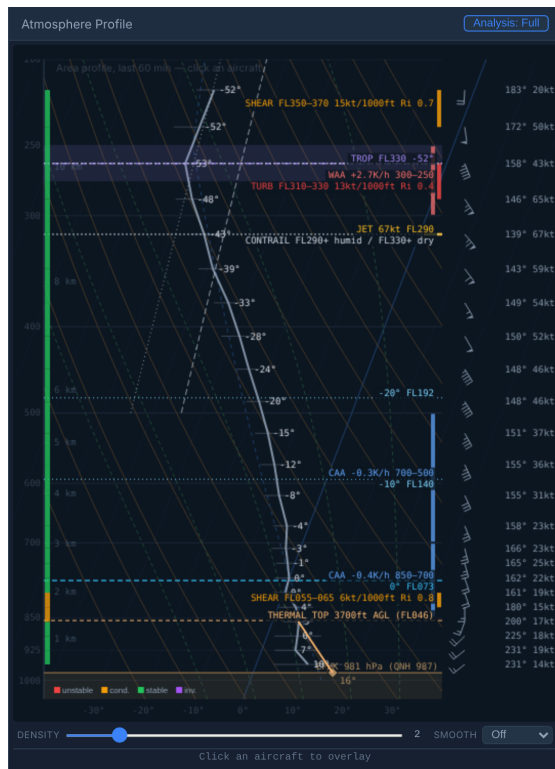
The shear must exceed twice the sampling uncertainty of the two mean winds, so random scatter alone does not trigger a flag. Typical places are just below and above the jet core, at the tropopause and in the lowest kilometre. Because the layers are 1 000–2 000 ft thick, thin shear layers are averaged out and real turbulence can be stronger and more local — treat the marks as indications, not as a turbulence forecast.

8.4 Advection and contrails

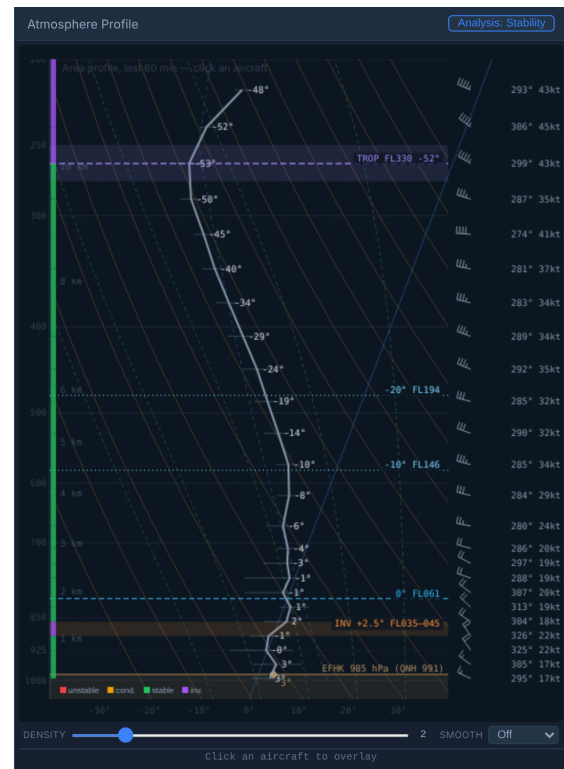
Warm and cold air advection (red / blue bar next to the shear bar, e.g. CAA 0.4K/h 850–700). In large-scale flow the change of wind with height (the *thermal wind*) is parallel to the isotherms of the layer. Wind turning clockwise with

height (veering) therefore means that warm air is being carried in, anticlockwise (backing) cold air. The rate in K per hour is estimated for the layers 850–700, 700–500, 500–400, 400–300 and 300–250 hPa. The layer below 850 hPa is left out because there friction turns the wind by itself. A layer is shown only when both winds are at least 10 kt, the wind turns at least 10° and the change exceeds its uncertainty. Expect the profile to warm in WAA layers and to cool in CAA layers over the next hours; in the archive the signs agreed with the observed temperature changes on 8–10 October 2026. Rates above about 1–2 K/h near the jet are usually exaggerated by wind noise.

Contrail levels (CONTRAIL FL290+ humid / FL330+ dry). The Schmidt–Appleman criterion gives the temperature below which an engine exhaust plume becomes saturated and forms a contrail. It depends on pressure and on the humidity of the ambient air. Two threshold curves are drawn above 500 hPa: dashed for air saturated with respect to water (about -42°C at 250 hPa), dotted for completely dry air (about -51°C). Where the profile lies left of the dashed curve, contrails form *if* the air is humid enough; left of the dotted curve they form at any humidity. Whether a contrail *persists* and spreads needs ice-supersaturated air, which cannot be seen without humidity data.



(a) Full with an illustrative surface temperature of 16°C : the dry adiabat from the surface (orange) meets the profile at THERMAL TOP 3700ft AGL (FL046).



(b) 10 October 2026, 04–05 UTC, Stability: after the low, cold air near the ground with an inversion INV $+2.5^\circ$ FL035–045, freezing level FL061, tropopause FL330.

Figure 6. Thermal top and an inversion case.

8.5 Full: daytime thermal top

The surface temperature of the METAR, at the station pressure, is lifted along the dry adiabat (orange line) until it is no longer warmer than the measured profile (Fig. 6a). That height is the top of dry thermals — the depth of the daytime mixing layer, important for glider pilots, smoke and pollutant dispersion and low-level turbulence. It is labelled e.g. THERMAL TOP 3700ft AGL (FL046). When the air just above the ground is already warmer than the surface air — at night and in the morning, or in cold cloudy weather — it shows THERMALS: none (stable surface layer). When it cannot be computed it shows THERMAL TOP: n/a with the reason (no METAR temperature yet, too few low-level data).

Without a dewpoint the *cloud base* is unknown, so cumulus may form below this top. No extra surface heating is added, so on sunny afternoons the real top is often a few hundred feet higher. The METAR is refreshed every 10 minutes; after a restart of the server, reload the page once the first METAR has arrived.

9 Limitations

No humidity. Aircraft transponders do not report humidity (the MRAR format can, but practically no aircraft send it). The diagram therefore has no dewpoint curve, and the following cannot be derived: cloud base and cloud layers, fog

risk, the lifting condensation level, convective energy (CAPE) and thunderstorm indices, whether icing actually occurs, contrail persistence, and the refractivity of the air (radio ducting). The moist adiabats are drawn as reference lines only.

Vertical resolution. A radiosonde reports every few metres; the area profile has layers of 1 000 ft (300 m) up to 10 000 ft and 2 000 ft (600 m) above. Thin features — a 20–50 m surface inversion, a sharp tropopause, a thin shear layer or a radio duct — are smoothed out or missed. Table 2 compares the automatic tropopause and freezing level with the Jokioinen soundings.

Table 2. Automatic analysis of the area profile against the full-resolution Jokioinen radiosonde (WMO 02963, about 100 km north-west of the receiver), pressure altitude.

Sounding	Tropopause, sonde / area	0 °C level, sonde / area
8 Oct 2026 12 UTC	11.4 / 11.3 km	1.6 / 2.2 km
9 Oct 2026 00 UTC	11.1 km / not found	2.4 / 2.1 km
9 Oct 2026 12 UTC	10.5 / 10.1 km	2.3 / 2.2 km
10 Oct 2026 00 UTC	9.9 / 9.4 km	0.8 / 1.6 km

Sampling. The profile exists only where aircraft fly. Cruise levels are well sampled by day, low levels mainly along the EFHK approach and departure paths, and the night hours (about 22–04 UTC) may have only a few aircraft. Layers need at least 3 replies; check the number of replies and the 10–90 % range in the hover read-out. The area is 150 km in radius, so the profile is an area average over an hour, not a point measurement at one moment — a front passing through the area blurs it.

Accuracy. Temperatures from Mach and TAS are good at cruise (about ± 1 K for layer means) but scatter by 3–4 K per reply at approach speeds; low-level layer means are therefore less certain, especially with few aircraft. Winds depend on correct heading, TAS and the absence of turns; they are good to a few knots at cruise and less reliable at low speed and in turbulence.

Heights. The vertical scale is pressure; the km markers and flight levels are pressure altitude. True heights differ with QNH and air temperature. Only the thermal top is given in feet above ground, computed hypsometrically.

Automatic analysis. The analysis uses simple, documented rules on averaged data. It can miss features, mark marginal ones, or be fooled by sparse data. Use it as a guide to the diagram, not as an official weather product; for flight planning use official forecasts and soundings.

10 Quick reference

Symbol	Meaning
grey line, values	area temperature profile, last 60 min, 150 km
coloured line, circle	selected aircraft and its latest point
thin horizontal bars	10–90 % range of the temperatures in a layer
dashed blue line	ISA standard atmosphere
bold blue diagonal	0 °C isotherm
faint brown solid lines	dry adiabats (every 10 K)
faint green dashed lines	moist (saturated) adiabats
brown line, shading, diamond	ground (station pressure from METAR QNH), surface temperature
violet band TROP	tropopause
dashed blue 0° / dotted –10°, –20°	isotherm levels
yellow tick JET / MAX / LLJ	maximum wind, jet stream, low-level jet
orange shading INV	inversion
strip at the pressure axis	stability: red unstable, amber conditional, green stable, violet inversion
red / amber bar TURB / SHEAR	turbulence risk / strong shear
red / blue bar WAA / CAA	warm / cold air advection (K/h)
dashed / dotted curves CONTRAIL	contrail thresholds for humid / dry air
orange line THERMAL TOP	dry adiabat from the surface and the thermal top

A Formulas

- Mach/TAS temperature: $T = 288.15 (TAS / (661.47 M))^2$ K; used for $Mach \geq 0.2$, $TAS \geq 60$ kt and within 40 K of ISA.
- Pressure of pressure altitude h (m): $p = 1013.25 (1 - 0.0065 h / 288.15)^{5.25588}$ hPa below 11 km.
- Station pressure: $p_{\text{stn}} = QNH (1 - 2.25577 \times 10^{-5} z)^{5.25588}$, z = airport elevation (m).

- Dry adiabat: $T = \theta (p/1000)^{R/c_p}$, $R/c_p = 0.2857$.
- Moist adiabat: $\frac{dT}{dp} = \frac{RT + Lr_s}{p(c_p + L^2 r_s \epsilon / (RT^2))}$, integrated from 1000 hPa; r_s saturation mixing ratio, $L = 2.501 \times 10^6$ J/kg, $\epsilon = 0.622$.
- Layer thickness (hypsothetic): $\Delta z = (R\bar{T}/g) \ln(p_1/p_2)$.
- Thermal advection: $-\vec{V} \cdot \nabla T = -\frac{f}{R \ln(p_1/p_2)} (u_1 v_2 - v_1 u_2)$, $f = 2\Omega \sin \phi$.
- Schmidt–Appleman: $G = \frac{EI_{H_2O} c_p p}{\epsilon Q (1 - \eta)}$ with $EI = 1.25$, $Q = 43.2$ MJ/kg, $\eta = 0.3$; $T_{LM} = -46.46 + 9.43 \ln(G - 0.053) + 0.72 \ln^2(G - 0.053)$ °C (saturated air), $T_{LC} = T_{LM} - e_s(T_{LM})/G$ (dry air).

Written with the assistance of Claude (Anthropic). Screenshots: MODE-S Wind, dark theme, with archived area profiles of 9 and 10 October 2026; surface values in Figs. 1–6a are illustrative METARs.